Trigonometric Functions and Circular Measure

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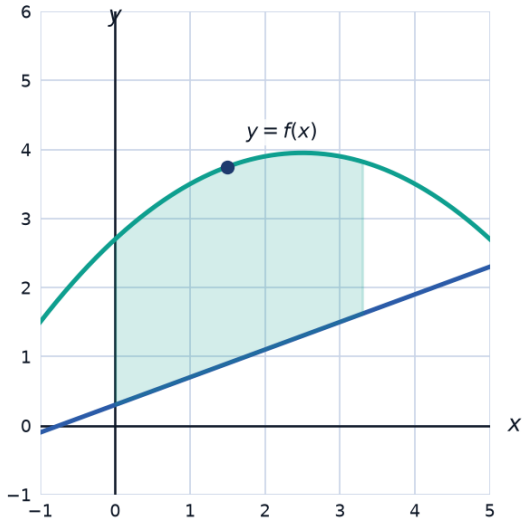
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

In a circle of radius $10\ \text{cm}$, chord AB subtends an angle of $\theta = \frac{2\pi}{3}$ radians at the centre O. Find: (a) the exact length of chord AB, (b) the exact area of the minor segment bounded by chord AB and minor arc AB, in terms of π and $\sqrt{3}$.

2 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle has radius $8\ \text{cm}$. A sector of this circle has a total perimeter of $28\ \text{cm}$. Find the angle of the sector in radians, and calculate the exact area of the sector.



3 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

4 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

5 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.

6 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

In ΔABC, $a = 8$, $b = 11$, and $A = 35^\circ$. Determine the number of possible triangles and find the measure of angle B for all valid cases to the nearest tenth of a degree.

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7 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

9 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

10 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A block of mass $5\ \text{kg}$ is released from rest at the top of a rough plane inclined at 30° to the horizontal. The length of the incline is $8\ \text{m}$ and the coefficient of friction is $\mu = 0.2$. Using the work-energy principle and taking $g = 9.8\ \text{m/s}^2$, calculate the speed of the block at the bottom of the incline.

11 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Determine whether the lines $\mathbf{r}_1 = (1; 2; -1) + \lambda(2; 1; 3)$ and $\mathbf{r}_2 = (0; 5; 4) + \mu(-1; 2; 1)$ intersect or are skew.

12 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

13 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A stone is thrown vertically upwards with speed $15\ \text{m/s}$ from the top of a cliff $40\ \text{m}$ above sea level. Taking $g = 9.8\ \text{m/s}^2$ and upwards as positive: (a) find the maximum height reached above the cliff, (b) calculate the time taken for the stone to reach the sea.

14 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Consider the points $A(1, 2, 3)$, $B(2, 4, 1)$, and $C(-1, 0, 5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

Find the binomial expansion of $\frac{1}{\sqrt{4-x}}$ in ascending powers of x up to and including the term in x^2. State the set of values of x for which the expansion is valid.

16 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Use the substitution $u = 2x + 1$ to find the exact value of $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx$.

17 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

A random variable X has distribution $\text{Po}(\lambda)$. A single observation of X is taken to test $H_0 : \lambda = 6$ against $H_1 : \lambda < 6$ at the 5% level. (a) Find the critical region. (b) State the actual significance level. (c) Find the probability of a Type II error if the true value of λ is 3.

18 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

A small particle of weight $24\ \text{N}$ is held in equilibrium by two light inextensible strings. String A makes an angle of 30° with the upward vertical, and string B makes an angle of 60° with the upward vertical. Find the tensions T_A and T_B in the two strings.

19 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ldots + (2n - 1) = n^2$.

20 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius \(r = 6 \text{ cm} \) and central angle \(\theta = \frac{\pi}{3} \text{ rad} \). What is the arc length of the sector?

a. \(\frac{6\pi}{3} \text{ cm} \)

b. \(\frac{36\pi}{6} \text{ cm} \)

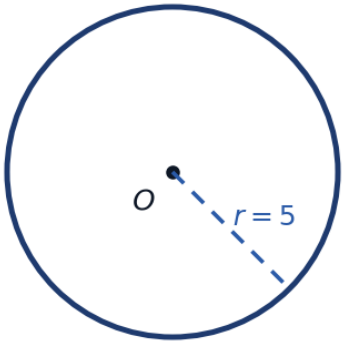
c. \(6\pi \text{ cm} \)

d. \(\frac{6\pi}{6} \text{ cm} \)

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22 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

The diagram shows a sector of a circle with centre O. The arc length of the sector is given by 6π cm and the area of the sector is 27π cm^2. The radius of the circle is r cm and the angle of the sector is θ radians. Find the value of r and θ.



a. r = 9, $\theta = \frac{2\pi}{3}$

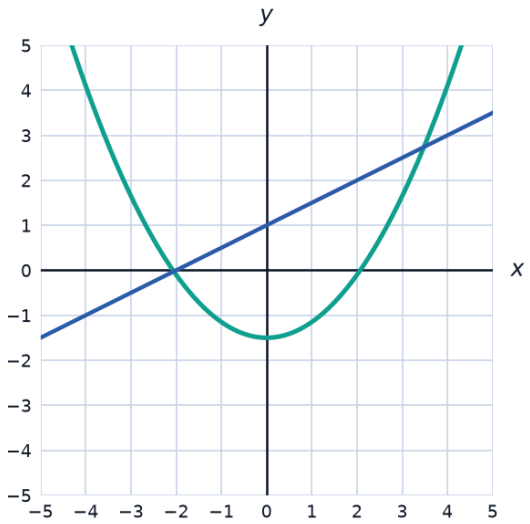
b. r = 6, $\theta = \pi$

c. r = 3, $\theta = 2\pi$

d. r = 18, $\theta = \frac{\pi}{3}$

23 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is 6 cm and the radius is 3 cm, find the angle θ in radians.



a. $\frac{1}{2}$ radians

b. 2 radians

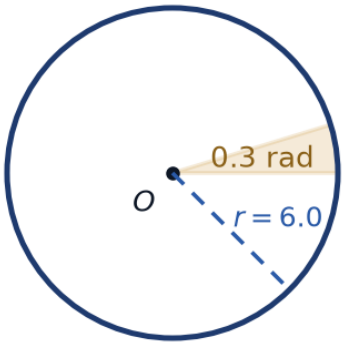
c. $\frac{2}{3}$ radians

d. $\frac{3}{2}$ radians

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24 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector has radius 6 cm and angle 0.3 rad. Find the arc length.



a. 9/5

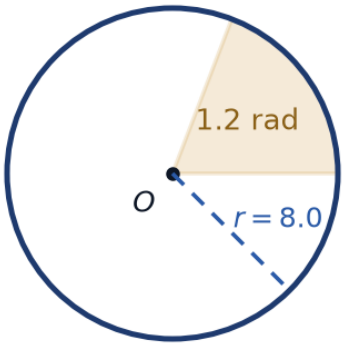
b. 63/10

c. 14/5

d. 27/5

25 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector OAB has radius 8 cm and angle $\angle AOB = 1.2$ rad. Find the area of the minor segment cut off by chord AB, leaving your answer in exact form.



a. $32(1.2 - \sin(1.2))$

b. $64(1.2 - \sin(1.2))$

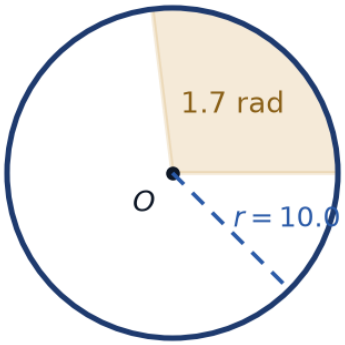
c. $33(1.2 - \sin(1.2))$

d. $32(1.2 + \sin(1.2))$

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26 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector has radius 10 cm and angle 1.7 rad. Find its area.



a. 170

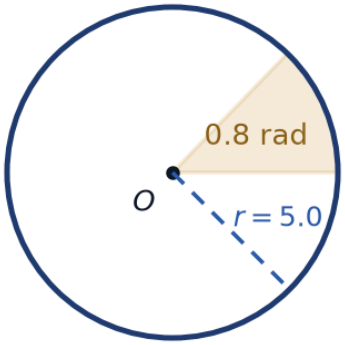
b. 95

c. 85

d. 17

27 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector has radius 5 cm and angle 0.8 rad. Find the perimeter of the sector.



a. 14

b. 9

c. 4

d. 15

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28 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector of a circle with radius r cm has a perimeter of P cm. The area of the sector is A cm^2. Prove that $P^2 \geq 16A$.

a. The inequality holds true as shown in the derivation.

b. The inequality holds true only for specific angles.

c. The inequality fails for angles greater than π radians.

d. The inequality fails for angles less than 2 radians.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

If $\sin\theta = -\frac{20}{29}$ and θ is in Quadrant 3, find $\cos\theta$.

a. $\frac{29}{21}$

b. $\frac{21}{29}$

c. $-\frac{21}{29}$

d. $\frac{20}{29}$

30 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Find the period and phase shift of $y = \tan(1/2x - 1)$.

a. Period 2π, phase shift 0

b. Period 2π, phase shift 2 units left

c. Period π, phase shift 2 units right

d. Period 2π, phase shift 2 units right

31 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Find the reference angle for 220°.

a. 60°

b. 40°

c. 50°

d. 140°

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32 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

What are the maximum and minimum values of $y = 3\sin(x)$?

a. Max 2, Min -2

b. Max 3, Min -3

c. Max 0, Min 0

d. Max 4, Min -4

33 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

What is the amplitude and period of $y = 5\sin(3x) - 3$?

a. Amplitude -3, period $\frac{2}{3}\pi$

b. Amplitude 5, period $\frac{2}{3}\pi$

c. Amplitude 5, period $\frac{3}{2}\pi$

d. Amplitude 5, period 2π

34 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Convert 150° to radians.

a. $\frac{5\pi}{6}$

b. $\frac{2\pi}{3}$

c. $\frac{\pi}{6}$

d. $\frac{7\pi}{4}$

35 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Convert 270° to radians, giving your answer in terms of π.

a. $\frac{\pi}{2}$

b. π

c. $\frac{3\pi}{2}$

d. $\frac{\pi}{6}$

36 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Convert $\frac{3\pi}{2}$ radians to degrees.

a. 30°

b. 180°

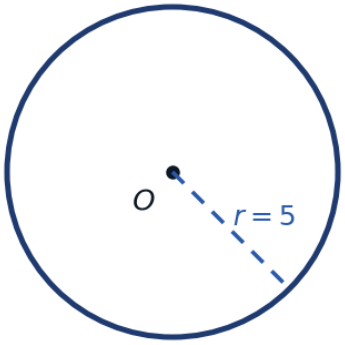
c. 90°

d. 270°

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37 **[Intermediate]** • **Target: Trigonometric Functions & Circular Measure Practice Exercises**

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.



a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

c. 1

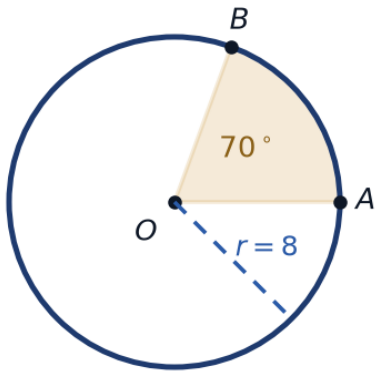
d. $\frac{\sqrt{2}}{2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?



a. $\cos\theta$

b. $\sin\theta$

c. $\tan\theta$

d. 1

39 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

40 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

What are the values of x that satisfy the equation $|3x - 2| = 2|x + 1|$?

a. $x = 0$ and $x = 4$

b. $x = 0$ and $x = -4$

c. $x = \frac{1}{5}$ and $x = 4$

d. $x = \frac{1}{4}$ and $x = \frac{3}{2}$

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41 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

a. 1

b. 2

c. 3

d. 4

42 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

The function f is defined for $x \in \mathbb{R}, x \neq 3$ by $f(x) = \frac{2x + 1}{x - 3}$. Which of the following is an expression for the inverse function $f^{-1}(x)$, where $x \neq 2$?

a. $\frac{3x + 1}{x - 2}$

b. $\frac{x - 3}{2x + 1}$

c. $\frac{3x - 1}{x + 2}$

d. $\frac{x + 1}{2x - 3}$

43 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

a. a = 5, b = 3

b. a = -5, b = 3

c. a = 5, b = 2

d. a = -5, b = 2

44 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

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45 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The diagram shows the graph of the function $y = f(x)$. The function $f(x)$ is defined for all real numbers.

Which of the following represents the graph of $y = f^{-1}(x)$?

a. Graph A

b. Graph B

c. Graph C

d. Graph D

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